Material	Description	Value	Quantity
Resistors	½ watt; Fixed	470 Ohm 680 Ohm 22 Kilo-Ohm 47 Kilo-Ohm 1.2 Kilo-Ohm	3 3 2 2 1
Buzzer	small	1.5 – 15V 10mA	1
Transistor	NPN	BC547	3
Power source	Battery/ DC power supply	4.5 – 12V	2
Diode	Light emitting diode	Red	2
Diode	Light emitting diode	Green	2
Diode	Light emitting diode	Yellow	2
PIR	Passive Infrared Sensor	HC-SR501	1
Switch	On/ Off switch	Contact switch	1
Mini pushbuttons (or Tactile Buttons)	Single pole-throw buttons	Mini pushbuttons	4
Capacitors	Electrolyte	10 micro-Farad 42 micro-Farad	2 2

3. Scenario and instructions for the tasks

Subtasks 1 to 4 (Fundamentals of programming)

In order to simulate the operations of a toll gate you are going to develop a Scratch Program as well as Python programs.

For the test scenario, a user will indicate what type of vehicles are passing the gate. The initial system will be tested with three types of vehicles and a fee per type as indicated in the table below.

Vehicle type	Code	Fee
Bike	B	R9.50
Car	C	R18.75
Trucks and, or (Busses)	T	R37.00

The total number of cars per type as well as the potential income for all the vehicles that are to pass the gate should be calculated for each day. Every day all counters and totals are to be reset to 0 after the day's calculation.

Subtasks 5 and 6 (EDCR)

In these subtasks you are going to build an electronic prototype for a toll gate alarm system and present the design as a breadboard prototype and in electronic form.

Subtask 7

In this subtask you will be required to print a small 3D artefact.

Subtasks 8 and 9

In these subtasks you are going to construct your 3D printed artefact and use it as part of the design of a prototype (Section 3.8 provides an overall high-level description of what is required)

3.1 **Subtask 1: IPO design and planning** (Total: 10 marks; 45 minutes)

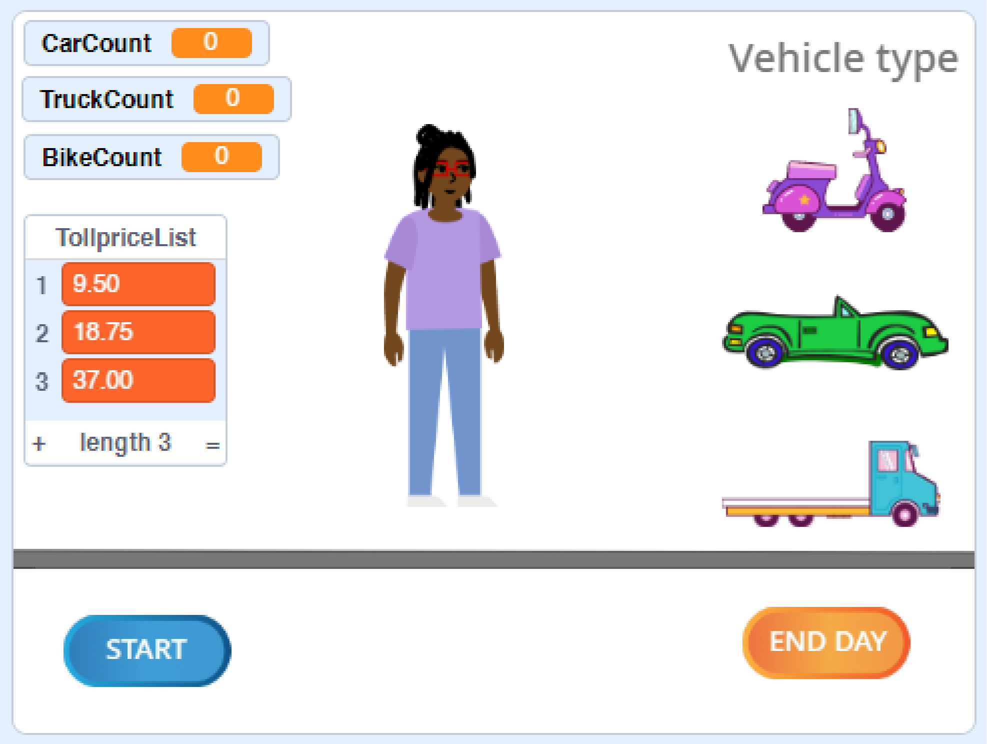
As part of the initial design of the system an IPO chart is required.

Your lecturer would have discussed the concept of IPO charting with you, and you would have completed several exercises of IPO charts already.

- Study the given scenario as presented in the problem statement for subtasks 1 to 4 (point 3) and design an IPO chart.
- Your first initial design can be on paper using a pen and a ruler.
- Under supervision use your initial design and construct an IPO chart using an appropriate software tool.
- Your IPO chart should include and indicate:
    - All inputs and outputs (as variables) as well as the suggested datatypes for each.
    - All intermediate variables.
    - High level algorithm as part of the process column.
- Add the lines of pseudocode below your IPO table to indicate how you are to calculate the total number of vehicles that will pass through the toll gate as well as the total potential income for the day.
- Your document should include your student details and PC details.
- Ensure that your IPO document are designed neatly and professionally.
- Submit your written copy of your initial IPO.
- Print your IPO and lines of pseudocode.

3.2 **Subtask 2: Simulation program in Scratch** (Total: 15 marks; 90 minutes)

Using Scratch, design a stage with similar (need not be the same) sprites as in the example below:

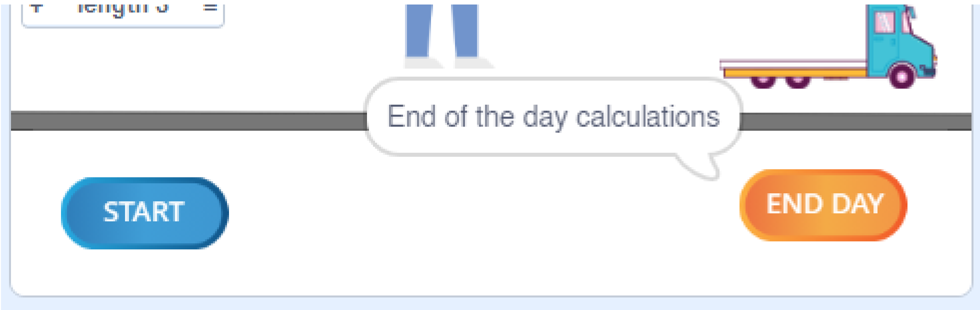


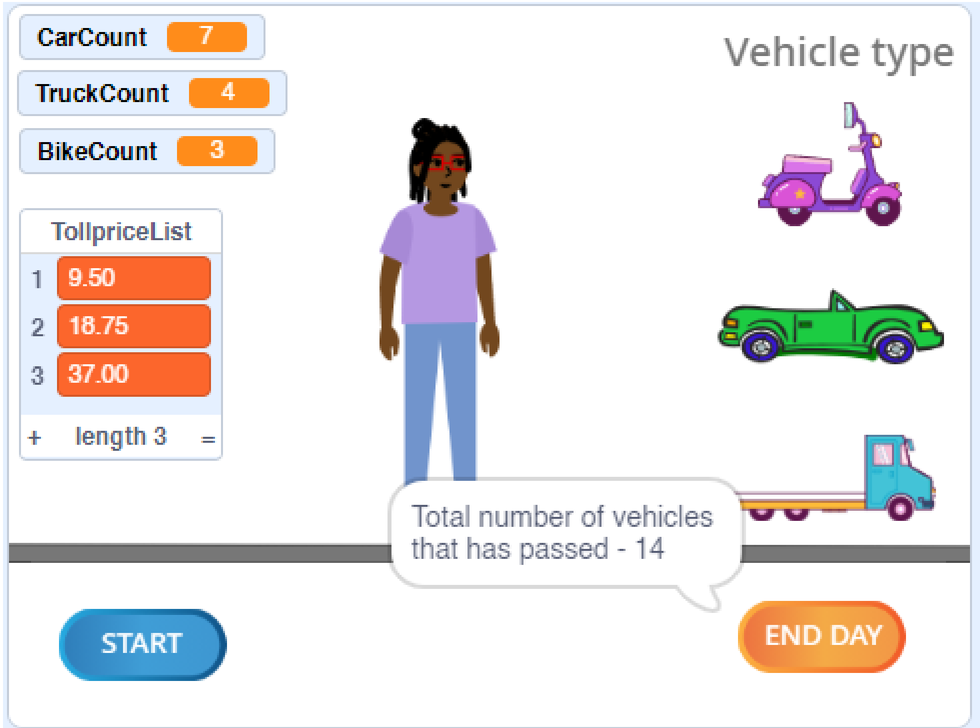
Remember to regularly save your program as you work, as part of your development process. (Name your program: TollGateApp)

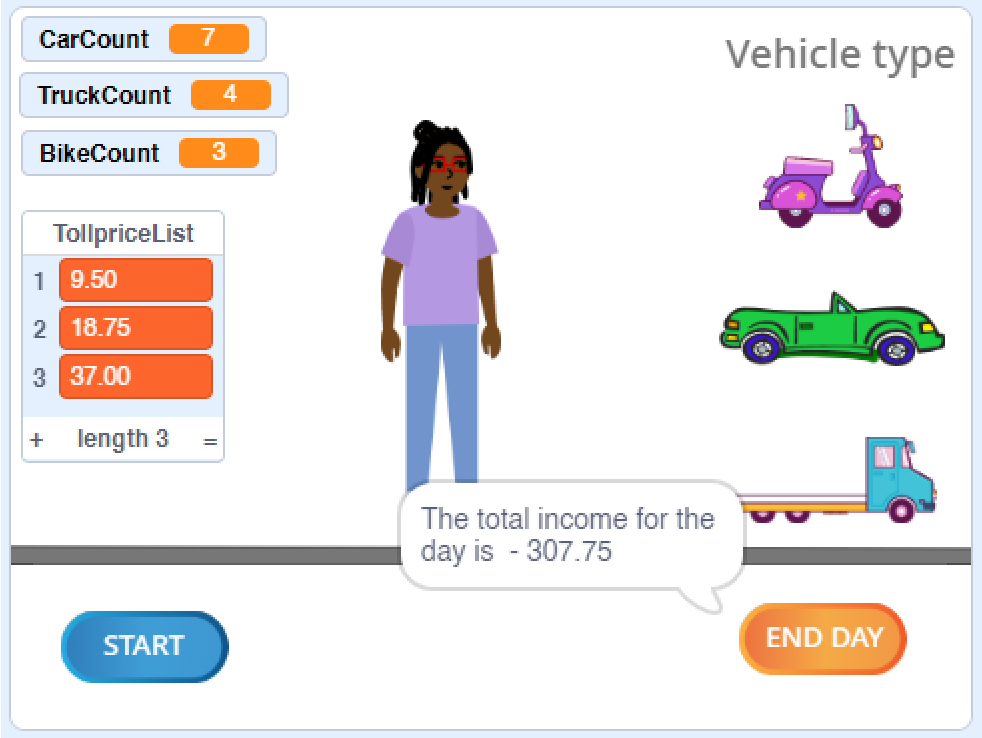
- Give all your sprites and the stage appropriate names.
- Write the code for the events and declare the applicable variables and a list that contains the prices per vehicle type. (Compare the list with the given table in 2.1).
- Write the code for the (START) sprite that will initialise all the variables to 0.
- Write the code that will when the user clicks on the (Bike, Car or Truck) sprites increment the respective counting variables.
- Declare all appropriate, intermediate variables.
- When the user for example, clicks on the bike, the operator (person sprite) should state the amount due for the particular vehicle.

- The counter for bikes is also incremented by 1.

- When the user clicks on the (END DAY) sprite a message should be displayed showing how many vehicles have passed and the total income.
- Study the following example output of a trial run.



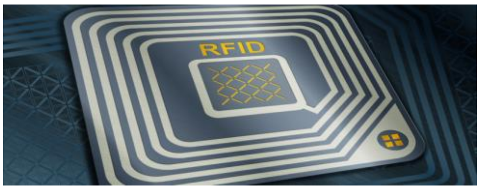




- Complete the Scratch program.
- Do a few sample, runs with your application.
- Save your program, by adding your student number to the file name.
- Compile a word document which includes your student details and the code included for each of the sprites.
- Take screenshots of the code of each of your sprites and compile a Word document with appropriate headings for each of the sprite codes. Add your code and add a sentence to explain what your code does.
- Print your document and submit both the document and the code to your lecturer.

3.3 **Subtask 3: Python program - String manipulation** (Total: 10 marks; 60 minutes)

In the future, the motor access control gate will be equipped with a feature that allows it to read (Radio Frequency Identification) RFID tags. This means that people who have an RFID tag will be able to gain access through the gate without having to physically stop and pay the fee. To make sure that only authorized individuals are granted access, you will need to create a program in the Python programming language. This program will be used to validate the RFID tag ID (Identification number) that is captured by the gate.



Write a Python program that prompts the user to enter an RFID tag and then validates the tag according to the specified format. If the RFID tag is valid, the program should display the message "RFID tag is valid." If the RFID tag is invalid, the program should display the message "RFID tag is invalid."